Karim Ehab

AI Engineer

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Summary

AI Engineer with strong software engineering skills, specializing in NLP, RAG systems, and AI agents using Python and FastAPI, with a focus on reliable, evaluation-driven, and explainable solutions. **Microsoft Certified: Azure AI Apps and Agents Developer Associate**, with hands-on experience building end-to-end RAG and agent-based systems.

Education

B.Sc. in Computer Engineering – Very Good with Honors Cairo, Egypt
Benha University – Faculty of Engineering at Shoubra

Experience

AI Engineer Intern Jul 2026 – Sep 2026
EVA Cosmetics *Cairo, Egypt*

- **Top 20 finalist among 1,500+ participants** in the EVA AI Hackathon, earning an AI Engineering internship.
- Developed an explainable AI-powered candidate screening and ranking system (see Projects), cutting screening time from **days to seconds** with transparent, traceable score breakdowns.
- Achieved **95.56% Precision@5** and **91.10% NDCG@10** on 150 CVs and 30 JDs, with 3–7s search latency.

NLP Trainee Jul 2025 – Aug 2025
National Telecommunications Institute (NTI) *Cairo, Egypt*

- Trained in NLP, LLMs, RAG, and Agentic AI; built sentiment analysis, OCR, translation, and chatbot applications using Hugging Face, crewAI, and LangChain.

Microsoft Machine Learning Engineer Trainee Apr 2024 – Oct 2024
Digital Egypt Pioneers Initiative (DEPI) *Cairo, Egypt*

- Led a 5-member team building a respiratory disease classifier across 6 categories using 26,000+ chest X-rays, achieving **93% accuracy** and ranking **Top 5 nationally** among DEPI ML track projects. **Received the Leadership Excellence award** as a top DEPI team leader.
- Trained in Machine Learning, Deep Learning, NLP, Computer Vision, Azure, and MLOps.

Projects

AI-Powered Candidate Search & Ranking System [Demo]
Python, FastAPI, LangChain, Qdrant, Redis, MongoDB, Gemini API, Docker, Azure

- Designed and implemented **explainable hybrid retrieval** with Qdrant, combining semantic and lexical search with recruiter metadata filters and weighted candidate scoring across skills, experience, and education.
- Designed a Dockerized microservices architecture with vendor-agnostic AI interfaces using the Factory Design Pattern, integrating CV parsing with LangChain, Gemini API, and Pydantic.
- Built a custom evaluation set of **150 CVs and 30 JDs** using LLM-assisted arena-style ranking to benchmark retrieval quality; added Redis caching to avoid repeated embedding and LLM calls.

AI First-Aid Assistant [GitHub]
Python, FastAPI, Qdrant, Docker, Pytest, Groq, Ragas

- Built a bilingual First-Aid RAG assistant with Qdrant hybrid dense/sparse retrieval fused via RRF, metadata filtering, and Groq Whisper voice queries; deployed as independent cloud microservices separating GPU-intensive workloads.
- Evaluated on 25 ground-truth queries, achieving approximately **98% Precision@3**, **100% Recall@3**, and **96% MRR**. Ragas evaluation reported **98% Answer Relevancy** and **100% Faithfulness**, with correct citations across all evaluated cases; validated using Pytest.

Technical Skills

LLMs & Agents: RAG, AI Agents, Function Calling, Multi-Agent Workflows, MCP, Prompt Engineering, LangChain, Hugging Face, OpenAI API, PyTorch, STT, TTS, A2A, Ragas

Retrieval & Vector Databases: Hybrid Retrieval, Metadata Filtering, Embeddings, Qdrant, Pinecone, FAISS

Backend & Languages: Python, SQL, FastAPI, Microservices, REST APIs, Pydantic, Design Patterns, Redis

Cloud & DevOps: Microsoft Azure, Azure AI Foundry, AI Search, Document Intelligence, Blob Storage, Docker, Git, GitHub

Leadership & Community

Head of AI — Google Developer Group on Campus, Benha University [Link] *Sep 2025 – Aug 2026*
Led AI workshops and hands-on projects for **30+ core members** across ML, NLP, LLMs, and Generative AI.

AI Content Creator — YouTube [Link] *Dec 2025 – Present*
Publishing technical content on AI, ML, DL, LLMs, RAG, vector databases, and production AI workflows.

Certifications & Achievements

- **Microsoft Certified: Azure AI Apps and Agents Developer Associate (AI-103)** (Oct 2026) — [Link]
- **Top 5 participants – Orange AI Hackathon (2026)** — AI First-Aid Assistant project — [Link]
- Retrieval-Augmented Generation (RAG) – DeepLearning.AI — [Link]